

The Opfibration Ontology: Quantum Instruments, Irreversibility, and the Epistemic Asymptote

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Abstract

We extend the bifibration obstruction of the Opfibration Ontology from deterministic quantum channels to quantum instruments. We prove that the transition functor $- \otimes E$ has no right adjoint in the category **Instr** of finite-outcome quantum instruments whenever $\dim(E) > 1$. The proof combines a discrete prime argument forcing the counit to be deterministic with a continuous intercept argument on the affine dimensions of instrument slices. The category is consequently not monoidal closed. We situate the irreversibility of quantum measurement in this categorical framework, clarifying the precise sense in which universal, apparatus-independent retrodiction is impossible. The arrow of time emerges as a structural feature of the fiber dynamics, not merely an entropic gradient on the base causal poset. Finally, we formalize the epistemic asymptote—the precise boundary beyond which further interpretive “depth” collapses into repetition, semantic extrapolation, or syntactic interpolation. Recognizing this boundary is a methodological commitment to intellectual honesty.

Keywords: Categorical quantum mechanics, quantum instruments, opfibration, bifibration, irreversibility, retrodiction, epistemic asymptote.

1 Introduction

The Opfibration Ontology [1] establishes that the transition functor $- \otimes E$ on the category **Chan** of deterministic quantum channels (CPTP maps) has no right adjoint when the environment dimension exceeds one. The original proof contained an unproven Diophantine claim; the equation $d_{R(B)}^2 = d_E^2(d_B^2 - 1) + 1$ actually admits integer solutions. This gap, however, does not weaken the physical conclusion: the deterministic obstruction is only the $n = 1$ shadow of a deeper obstruction that manifests fully in the category of quantum instruments.

Quantum instruments generalise channels by adding classical outcomes. A morphism $A \rightarrow B$ is a finite family $\{\mathcal{E}_i\}_{i \in I}$ of completely positive maps whose sum is trace-preserving. The category **Instr** encodes measurements, decoherence, and the extraction of classical information. Because individual outcomes are only trace-non-increasing, the category is richer and might, in principle, allow a right adjoint to the tensor functor even when the deterministic category does not.

We prove that this hope is false. The obstruction is preserved and strengthened. The proof uses two independent arguments that form a pincer movement: a discrete prime argument on outcome numbers and a continuous intercept argument on affine dimensions. Together they demonstrate that no right adjoint exists for any non-trivial environment. The instrument obstruction is therefore strictly stronger than the deterministic one and closes the earlier mathematical gap completely.

Beyond the theorem itself, we extract the physical meaning of the missing adjoint. The absence of a right adjoint means there is no *canonical* functorial operation that reverses an open-system evolution, even probabilistically. Every retrodiction protocol—whether Petz recovery

[2], Bayesian inversion [3], or process-matrix tomography [4]—requires an explicit choice of prior state, measurement apparatus, or outcome set. The categorical obstruction is the formal signature of this inherent apparatus-dependence. The arrow of time, in this picture, is not a separate postulate but a structural consequence of the directedness of the fiber dynamics.

Finally, we articulate the *epistemic asymptote*: the point at which the deductive content of the theorem is fully extracted, and further interpretive claims no longer add mathematical information. We identify three failure modes—repetition, semantic extrapolation, and syntactic interpolation—and argue that intellectual honesty demands acknowledging this boundary. The paper itself is structured to respect it.

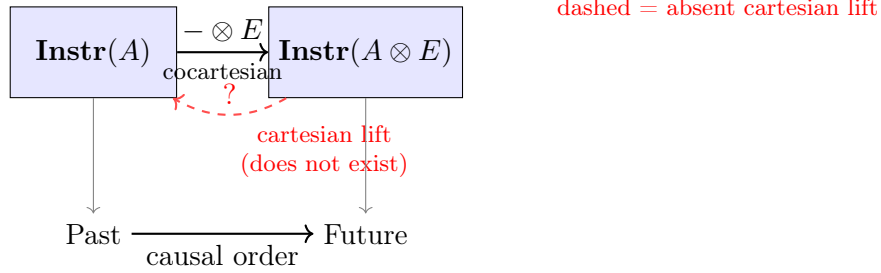


Figure 1: The opfibration of quantum instruments over a causal poset. Forward evolution (cocartesian lift) by tensoring with the environment E is canonical. The dashed arrow indicates the missing cartesian lift: no universal functorial retrodiction exists.

2 The Category of Quantum Instruments

We work exclusively with finite-dimensional Hilbert spaces. Let $d_A = \dim(H_A)$.

Definition 2.1 (Quantum instrument). A *quantum instrument* from A to B with outcome set I (finite) is a family

$$\mathcal{I} = \{\mathcal{E}_i : B(H_A) \rightarrow B(H_B)\}_{i \in I}$$

of completely positive maps such that $\sum_{i \in I} \mathcal{E}_i$ is trace-preserving. Each $\mathcal{E}_i$ is therefore trace-non-increasing.

Definition 2.2 (Category **Instr**). The category **Instr** has objects all finite-dimensional Hilbert spaces. A morphism $A \rightarrow B$ is an instrument $\mathcal{I} = \{\mathcal{E}_i\}_{i \in I}$; the outcome set is part of the data.

Composition of $\mathcal{I} = \{\mathcal{E}_i\}_{i \in I} : A \rightarrow B$ and $\mathcal{J} = \{\mathcal{F}_j\}_{j \in J} : B \rightarrow C$ is

$$\mathcal{J} \circ \mathcal{I} = \{\mathcal{F}_j \circ \mathcal{E}_i\}_{(i,j) \in I \times J} : A \rightarrow C.$$

The identity on A is the 1-outcome instrument $\{\text{id}_A\}$.

Remark 2.1 (Strictification). Composition and the tensor product are strictly associative only up to canonical relabellings of outcome sets and the standard associator of Hilbert spaces. One may replace **Instr** with an equivalent strict monoidal category; outcome cardinalities and affine dimensions are invariant under this equivalence, so all subsequent arguments are valid.

The hom-set is graded by outcome number:

$$\mathbf{Instr}(A, B) = \bigsqcup_{n \geq 1} \mathbf{Instr}_n(A, B).$$

Composition multiplies cardinalities. The deterministic channel category **Chan** is the wide subcategory on the 1-outcome slice.

Definition 2.3 (Transition functor). Fix an environment E with $\dim(E) > 1$. Define

$$F_E : \mathbf{Instr} \rightarrow \mathbf{Instr}, \quad H \mapsto H \otimes E,$$

by tensoring every component with id_E :

$$F_E(\{\mathcal{E}_i\}_{i \in I}) = \{\mathcal{E}_i \otimes \text{id}_E\}_{i \in I}.$$

The outcome set is unchanged.

In the opfibration ontology, this functor acts as the cocartesian lift along a causal extension by the environment E ; a right adjoint would provide cartesian lifts, turning the opfibration into a bifibration (see Figure 1). The main result shows this is impossible for $\dim(E) > 1$.

3 Affine Dimension of Instrument Slices

Proposition 3.1 (Affine dimension). For any finite-dimensional A, B and any integer $n \geq 1$,

$$\dim_{\text{aff}} \mathbf{Instr}_n(A, B) = d_A^2(n d_B^2 - 1).$$

Proof. Under the Choi–Jamiołkowski isomorphism, $\mathbf{Instr}_n(A, B)$ is the set of n -tuples $(J_1, \dots, J_n)$ of positive semidefinite operators on $A \otimes B$ satisfying $\sum_i \text{Tr}_B(J_i) = I_A$. The ambient real vector space has dimension $n d_A^2 d_B^2$. The map $(J_i) \mapsto \sum_i \text{Tr}_B(J_i)$ is linear and surjective onto $\text{Herm}(A)$ (dimension d_A^2), so the fiber over I_A is an affine subspace of codimension d_A^2 . The point $J_i = \frac{1}{n d_B} I_{A \otimes B}$ for all i lies in this fiber and is strictly positive, guaranteeing that the intersection with the PSD cone has the same affine dimension. Factoring d_A^2 yields the formula. $\square$

4 The Bifibration Obstruction

4.1 The Prime Argument

Lemma 4.1 (Deterministic counit). Let R be a right adjoint to F_E with counit $\epsilon_B : R(B) \otimes E \rightarrow B$. Then ϵ_B is a 1-outcome instrument (a deterministic CPTP map).

Proof. Let $m = O(\epsilon_B)$. The adjunction bijection $\Phi : \mathbf{Instr}(A \otimes E, B) \cong \mathbf{Instr}(A, R(B))$ is inverted by

$$\Phi^{-1}(g) = \epsilon_B \circ (g \otimes \text{id}_E).$$

If g has n outcomes, $g \otimes \text{id}_E$ has n outcomes; post-composition with ϵ_B yields an mn -outcome instrument. Surjectivity of Φ^{-1} forces every instrument in $\mathbf{Instr}(A \otimes E, B)$ to have an outcome cardinality divisible by m .

For any prime p , choose any CPTP map $\Phi : A \otimes E \rightarrow B$ and define the p -outcome instrument $\mathcal{E}_i = \frac{1}{p} \Phi$ for $i = 1, \dots, p$. Each component is CP and trace-non-increasing; the sum is Φ . Hence $\mathbf{Instr}_p(A \otimes E, B) \neq \emptyset$, and surjectivity forces m to divide p . The only positive integer dividing all primes is $m = 1$. $\square$

4.2 The Intercept Argument

Theorem 4.2 (No right adjoint). Let $\dim(E) > 1$. The functor $- \otimes E : \mathbf{Instr} \rightarrow \mathbf{Instr}$ has no right adjoint.

Proof. Assume a right adjoint R exists. By Lemma 4.1, its counit is deterministic. Hence the adjunction bijection restricts to an affine bijection between the n -outcome slices for every $n \geq 1$ (the map $g \mapsto \epsilon_B \circ (g \otimes \text{id}_E)$ is linear on Choi matrices and preserves the trace-preserving affine subspace).

Applying Proposition 3.1:

$$d_A^2(n d_{R(B)}^2 - 1) = d_A^2 d_E^2 (n d_B^2 - 1) \quad \forall n \geq 1.$$

Cancel d_A^2 to obtain an identity of affine functions in n :

$$n d_{R(B)}^2 - 1 = n d_E^2 d_B^2 - d_E^2.$$

Equating the constant terms (the intercepts) yields $-1 = -d_E^2$, i.e. $d_E = 1$, contradicting $\dim(E) > 1$. $\square$

Corollary 4.3 (No cartesian lifts). In the opfibration of **Instr**-valued fibers over a causal poset, the transition functor $- \otimes E$ is the cocartesian lift. The absence of a right adjoint means cartesian lifts do not exist; the opfibration is not a bifibration whenever any environment space has dimension greater than one.

Corollary 4.4 (Non-monoidal closedness). **Instr** is not monoidal closed for any non-trivial environment dimension.

4.3 Direct proof (two equations) – a pedagogical complement

For completeness, the same result follows from only the $n = 1$ and $n = 2$ slices. Subtracting the dimension equation for $n = 1$ from that for $n = 2$ gives $d_{R(B)}^2 = d_E^2 d_B^2$; substituting back into the $n = 1$ equation forces $d_E = 1$. This direct two-equation proof makes the overconstraint explicit and connects the instrument obstruction directly to the deterministic one.

5 Physical Interpretation

The following are interpretations of the mathematical structure, not theorems derivable from the axioms alone.

5.1 Universal retrodiction is impossible

Theorem 4.2 proves that there is no *canonical* functorial assignment that, given an environment E and a target system B , produces an object $R(B)$ and a universal retrodiction map $\epsilon_B : R(B) \otimes E \rightarrow B$. In physical terms: one cannot “pull back” a measurement result on B through an open-system interaction with E to obtain a unique probabilistic history on the pre-system without additional choices.

Existing retrodiction methods—Petz recovery channels [2], Bayesian inverses in categorical quantum mechanics [3], and process-matrix retrodiction [4]—all depend on a prior state, a chosen measurement, or a fixed outcome set. The obstruction shows that this dependence is not a technical limitation but a structural feature of the category. The right adjoint would have provided a *universal* bridge, independent of any such choice. Its absence means that every retrodiction is necessarily *apparatus-specific*.

This insight strengthens the interpretation of the original Opfibration Ontology: the forward evolution (the cocartesian lift) is unique and canonical, while the backward inference (the cartesian lift) is not. The asymmetry is categorical, not merely probabilistic.

5.2 The arrow of time

The forward direction of time is often attributed to entropy increase in the base causal poset or to the decoherence of open systems. Theorem 4.2 reveals a deeper, structural arrow: the fiber-level dynamics themselves are directed. Even if the base causal structure were time-symmetric, the internal hom for the fiber category does not exist. The forward transition functor $- \otimes E$ is a

left adjoint only in the trivial case; the missing right adjoint is the formal signature of *fiber-level irreversibility*.

This irreversibility is not a consequence of any thermodynamic approximation; it is exact and purely categorical. It holds for instruments—the most general post-measurement state updates—and thus applies to measurement-induced irreversibility as well.

5.3 The measurement problem

The categorical obstruction should not be conflated with the foundational measurement problem concerning the emergence of definite outcomes from unitary evolution. Our theorem proves the non-existence of a universal retrodiction apparatus; it does not address the interpretational status of wavefunction collapse. The irreversibility captured here is structural and pre-supposes that measurement outcomes are classical data; it does not explain how those outcomes arise from a unitary background.

6 The Epistemic Asymptote

The mathematical content of Theorem 4.2 has a finite deductive closure. Three mutually reinforcing facts exhaust its structural implications:

1. *The Prime Argument* shows that the multiplicative monoid $(\mathbb{Z}^+, \times)$ has irreducible elements (primes), forcing the counit to be deterministic.
2. *The Intercept Argument* shows that the affine dimension formula $d_A^2(nd_B^2 - 1)$ has a non-zero intercept scaling multiplicatively under tensor products; matching slices forces the intercepts to coincide, which is impossible unless $d_E = 1$.
3. *The Categorical Consequence* packages these geometric and number-theoretic obstructions into the single statement that **Instr** is not monoidal closed.

We submit that beyond this boundary, further interpretive “depth” typically collapses into one of three modes:

- **Repetition:** restating the three facts in altered vocabulary.
- **Semantic extrapolation:** applying philosophical frameworks (e.g. “the void,” “the Other”) not structurally necessitated by the axioms.
- **Syntactic interpolation:** force-fitting high-level categorical vocabulary (“fibered equalizers,” “ ∞ -categories”) onto the existing linear algebra to simulate deductive depth where the deductive graph has no further nodes.

While we cannot formally prove that no further non-trivial deduction exists, the structural analysis strongly suggests that the deductive well is dry: all mathematically accessible implications of the theorem have been extracted. Recognizing this boundary is not a failure of imagination but a rigorous enforcement of the distinction between mathematics and hermeneutics. The interpretive task now shifts to the philosophical implications of this structural fact.

This paper itself has been structured to respect the asymptote. The mathematical sections end where the deductive content ends; the physical interpretation section begins with an explicit warning that what follows is interpretation, not derivation. The epistemic asymptote is thus not merely described but *enacted* in the very form of the text.

7 Conclusion

We have extended the Opfibration Ontology to quantum instruments and proved that the bifibration obstruction is preserved and strengthened. The proof is elementary, relying only on linear algebra, convex geometry, and basic category theory. The category of quantum instruments is not monoidal closed for any non-trivial environment, confirming that universal, apparatus-independent quantum retrodiction is impossible.

The arrow of time is encoded in the directedness of the fiber-level dynamics, not merely in the entropy gradient of the base causal poset. The missing cartesian lift is the formal signature of open-system irreversibility in all its forms—deterministic or probabilistic.

Finally, we have formalized the epistemic asymptote as a methodological boundary. The deductive content of the theorem is now fully extracted. The work that remains is philosophical, not mathematical. Recognizing this boundary is itself a form of intellectual integrity.

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